Systems of rational inequalities

Solve each one alone, draw them on the same line, and keep only the overlap.

LESSON 35–37 | 3 × 40 MIN | ALGEBRA 10, §2.4 | P1 · 1.2 (EXTENSION)

LESSON CLOCK

10 | 22' | 26' | 24' | 38' | 15'

And versus or	10 min
The method	22 min
Worked systems	26 min
Empty and unbounded answers	24 min
Practice	38 min
Homework	15 min

LEARNING OBJECTIVES

- Solve each inequality of a system separately.
- Represent the solution sets on one number line and read the intersection.
- Distinguish a system (“and”) from a collection (“or”).
- Handle systems whose solution set is empty.

TEXTBOOK

National	pp. 129–142
Cambridge	Pure Mathematics 1, pp. 6–11
Workbook	P1 Exercise 1C

01 Explanation

“And” or “or”

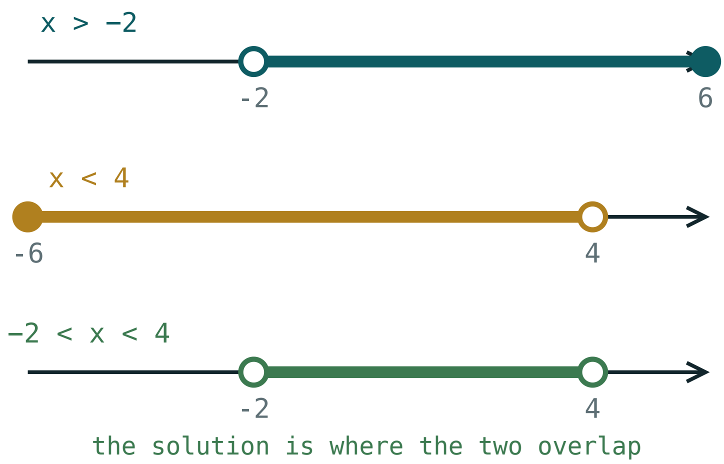
TWO DIFFERENT QUESTIONS

A **system** — written with a brace — asks for the values satisfying **every** inequality: the **intersection**. A **collection** asks for those satisfying **at least one**: the **union**.

$x > 1$ and $x < 4$ gives $1 < x < 4$. The same two as a collection give all of $\mathbb{R}$. The brace matters.

The method

1. Solve each inequality on its own, by the sign-chart method of the last lesson.
2. Draw all the solution sets on **one** number line, one above another.
3. Shade the part covered by every line.
4. Read the answer as an interval or a union of intervals.



Three solution sets drawn above one another. The answer is the strip covered by all three.

> DO NOT TRY TO SOLVE THEM SIMULTANEOUSLY
>
> Inequalities are not equations: adding or subtracting them loses information. Solve each separately, then intersect on the picture.

Worked systems

Example. $\frac{1}{x} > 0$ and $\frac{x-2}{x+1} \le 0$.

INEQUALITY	SOLUTION
$\frac{1}{x} > 0$	$x > 0$
$\frac{x-2}{x+1} \le 0$	$-1 < x \le 2$
intersection	$0 < x \le 2$

A double inequality such as $-3 < \frac{2}{x} < 1$ is a system in disguise: break it into $\frac{2}{x} > -3$ and $\frac{2}{x} < 1$, solve both, intersect. It cannot be solved “all at once” because the middle contains x.

When the answer is empty

$x > 5$ and $x < 2$ have no common value. That is a legitimate answer, written $\varnothing$ or “no solution”, and it must be justified by the picture — not left blank.

THREE SHAPES OF ANSWER

a single interval · a union of intervals · the empty set. Anything else means an error in one of the individual solutions.

Systems are the natural home of the **domain** conditions from Quarter I. "Find the domain of $\frac{\sqrt{x-1}}{x-4}$" is exactly the system $x - 1 \geq 0$ and $x \neq 4$.

02 Worked examples

EXAMPLE 1 **Solve the system $2x - 1 > 3$ and $\frac{1}{x-5} < 0$.**

1. First: $x > 2$.
2. Second: $x < 5$.

   The numerator is positive.
3. Intersect.

ANSWER

$2 < x < 5$

EXAMPLE 2 **Solve $\frac{x}{x-3} \geq 0$ and $x^2 - 16 < 0$.**

1. First: $x \leq 0$ or $x > 3$.
2. Second: $-4 < x < 4$.
3. Intersect both pieces.

ANSWER

$-4 < x \leq 0$ or $3 < x < 4$

EXAMPLE 3 **Find the domain of $f(x) = \frac{\sqrt{x+2}}{x^2-9}$.**

1. Radicand ≥ 0: $x \geq -2$.
2. Denominator $\neq 0$: $x \neq \pm 3$.
3. $x = -3$ is already excluded by the first.

ANSWER

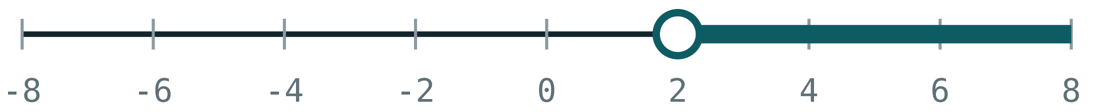
$x \geq -2, x \neq 3$

03 Interactive model

Draw both sets on the board one above the other; the answer is visible before it is written.

Intersecting two solution sets

$x > 2$



inequality `x > 2` | boundary `2 (open)` | x = 0 works? **no** | x = 6 works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
System of inequalities	Tengsizliklar sistemasi	Система неравенств
Collection of inequalities	Tengsizliklar majmuasi	Совокупность неравенств
Intersection of sets	To‘plamlar kesishmasi	Пересечение множеств
Union of sets	To‘plamlar birlashmasi	Объединение множеств
Empty solution set	Bo‘sh yechimlar to‘plami	Пустое множество решений
Number line	Sonlar o‘qi	Числовая прямая
Common part	Umumiy qism	Общая часть
Double inequality	Qo‘sh tengsizlik	Двойное неравенство
Bounded interval	Chegaralangan oraliq	Ограниченный промежуток

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A system of inequalities asks for:

- the union
- the intersection
- either one
- the difference

$x > 1$ and $x < 4$ gives:

- $1 < x < 4$
- $\mathbb{R}$
- $x > 4$
- $\varnothing$

$x > 5$ and $x < 2$ gives:

- $2 < x < 5$
- all reals
- $\varnothing$
- $x > 5$

$-3 < \frac{2}{x} < 1$ should be:

- solved at once
- split into two inequalities
- cross-multiplied
- squared

The domain of $\frac{\sqrt{x}}{x-1}$ is:

- $x \ge 0$
- $x \ge 0, x \ne 1$
- $x > 1$
- $\mathbb{R}$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Solve $x > 2$ and $x < 7$

   ANSWER $2 < x < 7$

2. Solve $x \geq 0$ and $x \leq 5$

   ANSWER $0 \leq x \leq 5$

3. Solve $x > 4$ and $x < 1$

   ANSWER $\varnothing$

4. Solve $2x > 6$ and $x - 1 < 4$

   ANSWER $3 < x < 5$

5. Domain of $\dfrac{\sqrt{x}}{x - 2}$

   ANSWER $x \geq 0, x \neq 2$

6. Solve $x^2 < 9$

   ANSWER $-3 < x < 3$

7. Solve $x > 0$ and $x^2 < 4$

   ANSWER $0 < x < 2$

Medium · the standard question

7 PROBLEMS

1. Solve $2x - 1 > 3$ and $\frac{1}{x - 5} < 0$

ANSWER $2 < x < 5$

2. Solve $\frac{x}{x - 3} \geq 0$ and $x^2 - 16 < 0$

ANSWER $-4 < x \leq 0$ or $3 < x < 4$

3. Solve $-3 < \frac{2}{x} < 1$

ANSWER $x < -\frac{2}{3}$ or $x > 2$

4. Domain of $\frac{\sqrt{x + 2}}{x^2 - 9}$

ANSWER $x \geq -2, x \neq 3$

5. Solve $\frac{1}{x} > 1$ and $x > 0.5$

ANSWER $0.5 < x < 1$

6. Solve $x^2 - 5x + 6 \leq 0$ and $x \neq 2$

ANSWER $2 < x \leq 3$

7. Solve $\frac{x - 1}{x + 2} > 0$ and $x < 3$

ANSWER $x < -2$ or $1 < x < 3$

Hard · stretch and reason

7 PROBLEMS

1. Solve $\frac{1}{x-1} > 2$ and $\frac{1}{x} < 3$

ANSWER $\frac{1}{3} < x < 1.5$ — check ends

2. Solve $x^2 - 4x < 0$ and $\frac{x-3}{x} \geq 0$

ANSWER $3 \leq x < 4$

3. Domain of $\frac{\sqrt{4-x^2}}{x-1}$

ANSWER $-2 \leq x \leq 2, x \neq 1$

4. Domain of $\sqrt{\frac{x-1}{x+2}}$

ANSWER $x < -2$ or $x \geq 1$

5. Find all k for which $x > k$ and $x < 2k$ is non-empty

ANSWER $k > 0$

6. Solve $|x-2| < 3$ and $\frac{1}{x} > 0$

ANSWER $0 < x < 5$

7. Solve $\frac{x^2-1}{x} \geq 0$ and $x \leq 2$

ANSWER $-1 \leq x < 0$ or $1 \leq x \leq 2$

07 Homework

Homework — 6 tasks

Draw one number line per system, with every solution set on it.

1. Solve $3x + 1 > 7$ and $\frac{1}{x-6} < 0$.

2. Solve $\frac{x + 1}{x - 2} \le 0$ and $x > -3$.

3. Solve $-2 < \frac{3}{x} < 1$.

4. Find the domain of $\frac{\sqrt{x - 1}}{x^2 - 16}$.

5. Find the domain of $\sqrt{\frac{x + 3}{x - 1}}$.

6. Give a system of two inequalities whose solution set is empty, and prove it with a number line.